

24/09/2025

Task-6-1

utilizing 'Functions' concepts in python programming.

Aim: To write the python programming using 'Functions' concepts in python program.

Algorithm:-

- 1) start the program.
- 2) print a welcome message
- 3) Determine and print the number of students.
- 4) print the type of lists:
- 5) Find and print highest and lowest grades.
- 6) print sorted list of grades.
- 7) stop

program:-

```
def analyze-student-grades():  
    #sample data  
    student_names = ["Alice", "Bob", "Charlie", "Diana"]  
    student_grades = [85, 92, 78, 90]  
  
    # 1. print a welcome message  
    print("welcome to the student!\n")  
  
    # 2. Determine and print the number  
    num_students = len(student_names)  
    print("Number of students:")  
  
    # 3. print the type of the student names list  
    print("\n Type of student - names list:",  
  
    # 4. Find and print and highest and lowest grade  
    highest_grade = max(student_grades)  
    lowest_grade = min(student_grades)  
    print("\n Highest grade:", highest_grade)  
    print("Lowest grade:", lowest_grade)  
  
    # 5. print the list of grades sorted in ascending order  
    sorted_grades = list  
  
    # 6. print the list of grades in reverse order  
    reversed_grades = list
```

Output:

Welcome to the students grades analyzer!

No. of students = 4

Type of students_names list: <class 'list'>

Type of students_grades list: <class 'list'>

Highest grade = 92

Lowest grade = 78

Sorted grades: [78, 85, 90, 92]

Reversed grades: [92, 90, 85, 78]

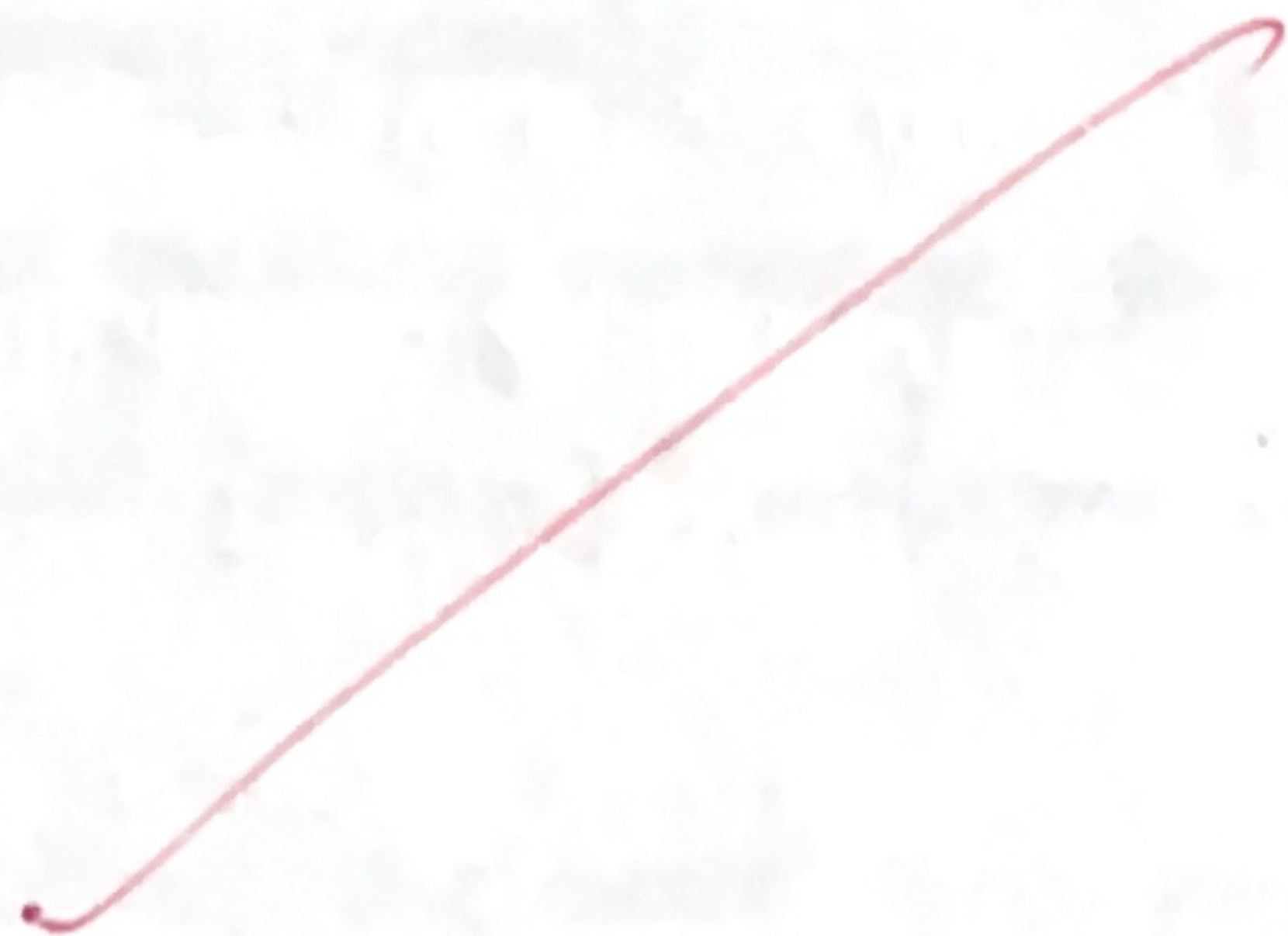
Grade indices from 1 to number of students: [1, 2, 3, 4]

#7 Generate and print a range of grade indices
from 1 to the number.

print("\n Grade indices from 1 to number of students:",
grade-indices)

Run the analysis.

analyze-student-grades()



Task-6.2

Aim: It is to create a small calculator application to help users perform basic arithmetic operations.

Algorithm:

- 1) start the program.
- 2) user input for Numbers.
- 3) user input for operation.
- 4) perform operation.
- 5) Display Result.
- 6) stop.

Program:

```
def add(a,b):  
    """Return the sum of two numbers."""  
    return a+b  
def subtract(a,b):  
    """Return the difference between two numbers."""  
    return a-b  
def multiply(a,b):  
    """Return the product of two numbers."""  
    return a*b  
def divide(a,b):  
    """Return the quotient of two numbers. Handles.  
    division by zero."""  
    if b!=0  
        return a/b  
    else:  
        return "Error: Division by zero"  
def greet(name):  
    """Return a greeting message for the user."""  
    return f"Hello, {name}! welcome to the program."  
def main():  
    # Demonstrating the user of user-defined functions.  
    # Arithmetic Operations  
    num1 = 10  
    num2 = 5  
    print("Arithmetic Operations!")
```

Output:

Arithmetic Operations

Sum of 10 and 5: 15

Difference between 10 and 5: 5

Product of 10 and 5: 50

Quotient of 10 and 5: 2

Greeting:

Hello, Alice! welcome to the program

```
#Greeting the user.
```

```
user_name = "Alice"
```

```
print ("\n Greeting!")
```

```
print (greet(user_name))
```

```
# Run the main function.
```

```
if __name__ == '__main__':
```

```
main()
```

1
2
3
4
5
6
7
8
9
10
11
12

VEL TECH	
EX NO.	6
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	
TOTAL (20)	
DATE WITH DATE	15

Result: Thus, the python program using 'Functions' concepts was successfully executed and the output was verified.